M.Tech Program

Advanced Industry Integrated Programs

Jointly offered by University and LTI MindTree

Graph RAG based Financial Insights

Review 3

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Introduction…. Calibri body -36

Problem

- Annual reports = **Tables + Text + Strategy + Risk**
- Financial information is **distributed across pages**
- Traditional search finds **text**, not relationships
- LLMs may generate **unsupported answers**

Proposed Idea

- Financial Fact Extraction
- Entity & Relationship Extraction
- **Financial Knowledge Graph**
- Multi-Hop Graph Retrieval
- **API or Local Model**
- Source/Page Grounding

Objective

- Extract **Sinancial facts**
- Identify **Susiness entities**
- Discover **Semantic relationships**
- Build a **Financial knowledge graph**
- Perform **Multi-hop reasoning**
- Retrieve **Source/page evidence**
- Generate answers using **LLM (API) local model**
- Minimize **Hallucination**

Module Name : Fin Graph

- Data Acquisition & Ingestion
- Document Parser
- Knowledge Graph Builder
- Intelligent Query Router
- Cold-Start Engine
- Multi-Hop Graph Reasoning, Evidence & Provenance Layer
- LLM Answer Generator
- Hallucination Guard

Problem Statement (Why fin Graph is better)

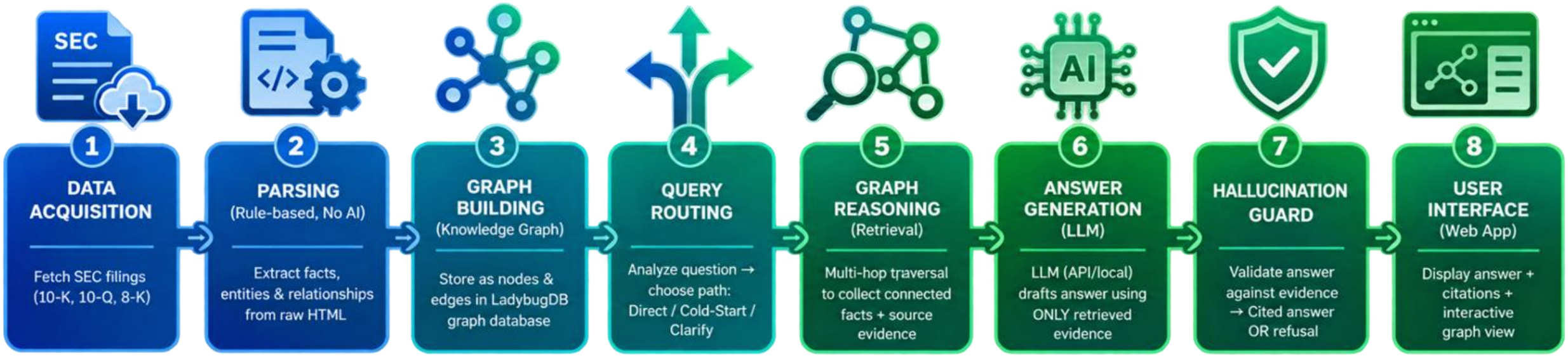
Every answer has proof : Each answer links back to the original SEC filing.

No fake data : Answers from real evidence — if data is missing, it denies.

Connects the dots : Graph reasoning links facts across companies and years to answer complex questions.

All-in-one platform : Ingestion, knowledge graph, reasoning, and AI Q&A — everything in one system.

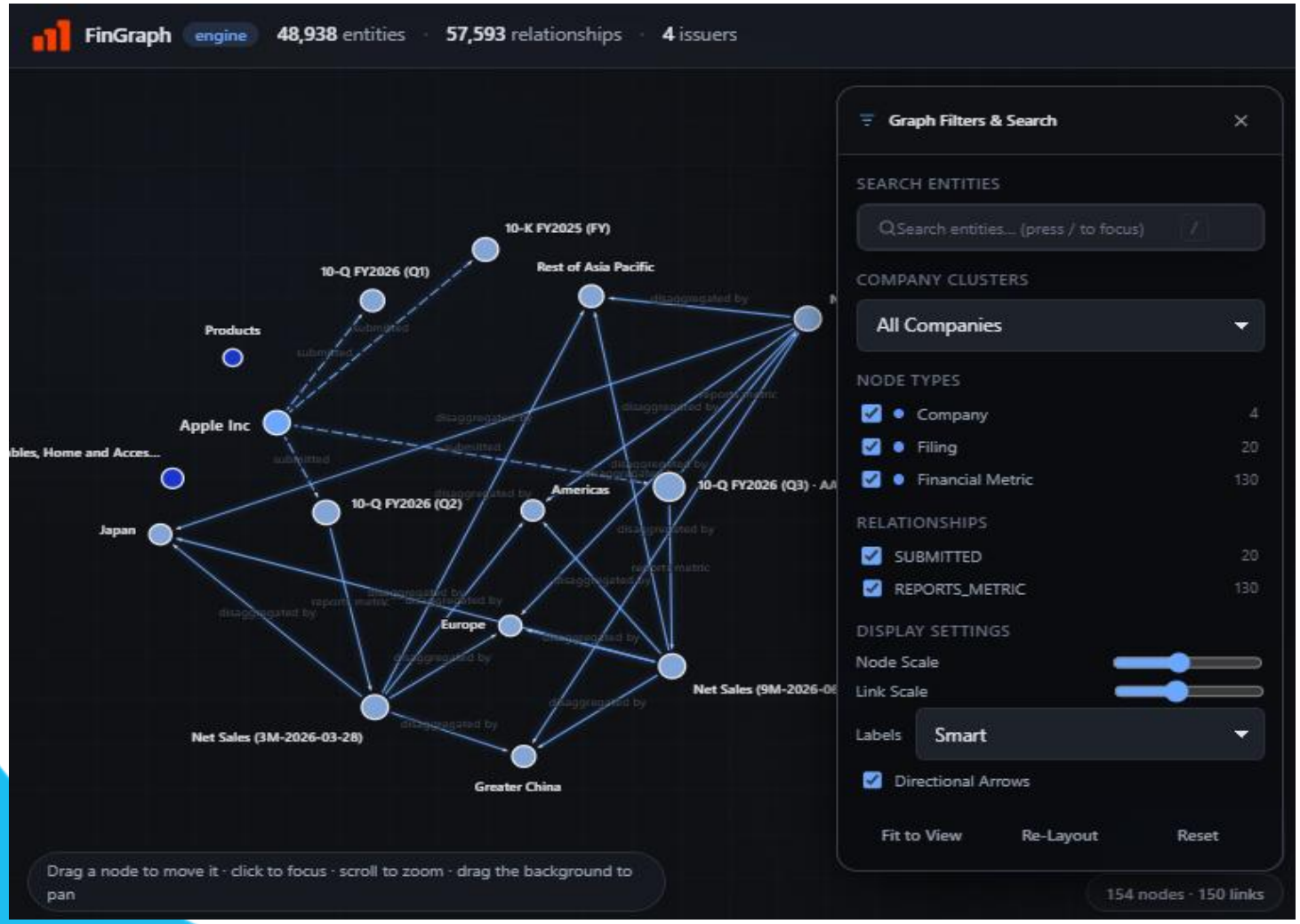
Methodology

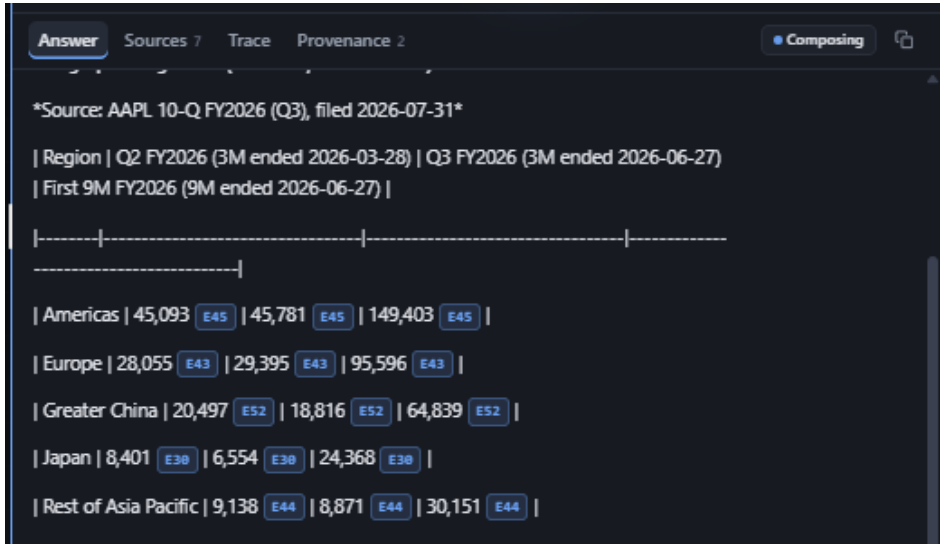


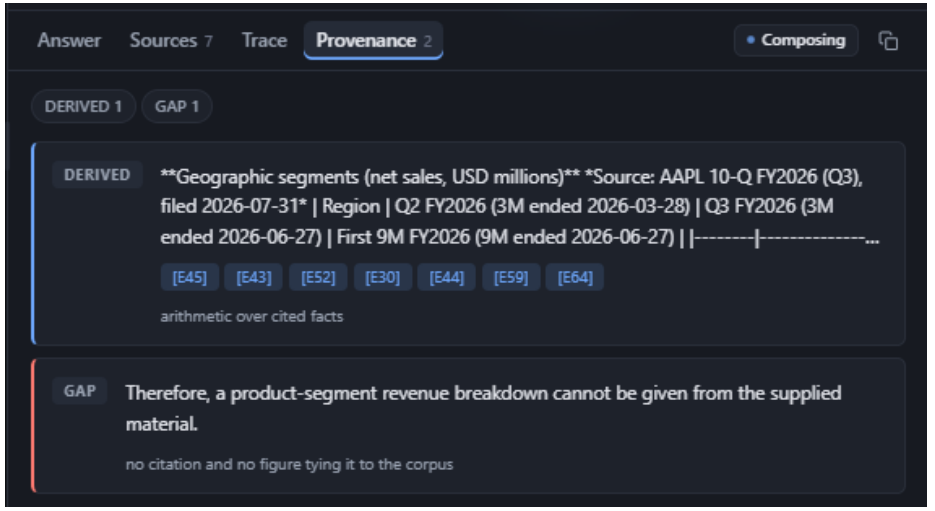
Approach/Algorithms/Tools

Step	Approach / Algorithm	Tools / Technologies
Data Acquisition & Parsing	Automated fetching of SEC filings + rule-based parsing (zero-LLM) to extract facts & entities	Python, SEC EDGAR API, BeautifulSoup
Knowledge Graph Construction	Property graph model — entities as nodes, relationships as edges, idempotent MERGE writes	LadybugDB, Cypher
Query Routing & Cold-Start	Classifies query path (Direct / Cold-Start / Clarify); LLM extraction for unseen companies	Python, LLM API
Multi-Hop Graph Retrieval	Bounded graph traversal (BFS) with company scoping to collect connected facts	Cypher, Python
Evidence & Provenance	Tags every fact — STATED / DERIVED / INFERRED / EXTERNAL / GAP with source/page citation	Python, retrieval layer
Answer Generation & Hallucination Guard	LLM generates answer only from evidence; grader validates figures, blocks unsupported claims	OpenAI / Gemini API, Python grader
Validation & Testing	Benchmarks, anti-leak tests, latency & security checks	Pytest, golden queries

Findings and Results







Conclusion… Calibri body -36

Achievements

- Developed an end-to-end system that converts SEC filings into a queryable financial knowledge graph
- Delivered accurate answers with source citations and zero hallucinated financial figures
- Ensured strict company isolation — no cross-company data mixing in results

Future Work

- Build graph segments on-the-fly during query time instead of pre-loading all filings
- Auto-process new filings daily so the system stays up-to-date without manual runs
- Reduce storage and startup time by loading only the data each query needs

Refrenences

Paper	Key Contribution
Lewis et al. (2020) — RAG	External knowledge retrieval + LLM generation
Reimers & Gurevych (2019) — SBERT	Semantic embeddings for similarity search
Chen et al. (2021) — FinQA	Numerical reasoning over financial reports
Zhu et al. (2021) — TAT-QA	Question answering over financial text + tables
Edge et al. (2024) — GraphRAG	Knowledge graphs + multi-hop/global reasoning